



Indium (In)- and tin (Sn)-based metal induced crystallization (MIC) on amorphous germanium (α -Ge)



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ABSTRACT

In this paper, metal-induced crystallization (MIC) phenomenon on α -Ge by indium (In) and tin (Sn) are thoroughly investigated. In- and Sn-MIC process respectively started at 250 °C and 400 °C. Compared to the previously reported MIC samples including In-MIC, Sn-MIC process presented higher sheet resistance (similar to that of SPC) and bigger crystal grains above 50 nm (slightly smaller than that of SPC). According to SIMS analysis, Sn atoms diffused more slowly into Ge than In at 400 °C, providing lower density of heterogeneous nuclei induced by metals and consequently larger crystal grains.

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1. Introduction

Germanium (Ge) possesses higher carrier mobility and lower fabrication process temperature than silicon (Si), making it an attractive channel material for monolithic three-dimensional integrated circuits (3D-ICs). For the 3D-ICs, both low temperature crystallization and activation processes are required to prevent degradation of existing copper (Cu) and aluminum (Al) interconnect layers. Therefore, various process techniques have been developed to crystallize Ge on an insulating layer and activate dopants in Ge at below 400 °C. Up to present, metal-induced crystallization (MIC) [1], laser annealing (LA) [2], boron (B) seed-based chemical vapor deposition (CVD) [3], and layer exchange (LE) [4] techniques have been reported to obtain polycrystalline Ge on an insulating layer at a low temperature regime. Especially, MIC and LA-based processes were also explored to activate dopants in Ge. Among these techniques, MIC has been widely researched because it is applicable to both crystallization and activation processes in large wafer-scale, making it more feasible for industry adoption. This technique uses metals atoms to create heterogeneous nuclei in Ge at very low temperatures, where Ge can grow as crystal grains. In the previous works, various metals such as nickel (Ni), gold (Au), aluminum (Al), palladium (Pd), Copper (Cu), cobalt (Co), platinum (Pt), and titanium (Ti) which exhibited melting temperatures between 660 °C (for Al) and 1772 °C (for Pt) were used to study the MIC phenomenon. For these

materials, MIC triggering temperatures were found to be in between 250 °C and 350 °C [5–9], which meant that the metals would not evaporate during the MIC process. However, MIC phenomena on Ge by indium (In) and tin (Sn)-like metals with very low melting temperatures (156 °C for In and 232 °C for Sn) were not reported yet because they are expected to re-evaporate during the MIC process. In this paper, In- and Sn-based MIC phenomenon on amorphous (α)-Ge is newly reported and compared with the previously reported MIC processes. Because of their low melting temperatures, it is important to prevent the metals from being re-evaporated and agglomerated during the MIC process. To prevent this, 100 nm thick SiO₂ was deposited on top of In and Sn. In addition, we experimentally investigate the properties of poly-Ge films crystallized by In- and Sn-MIC processes through X-ray diffraction (XRD), 4-point probe, transmission electron microscope (TEM), secondary ion mass spectroscopy (SIMS), and transfer length method (TLM) measurements.

2. Experiments

About 90 nm thick pure α -Ge film was deposited on 100 nm thick SiO₂ layer grown on Si(100) substrate by e-beam evaporating system. For the fabrication of In-MIC and Sn-MIC samples, 5 nm thick In or Sn layer was deposited on the α -Ge and then 100 nm thick SiO₂ capping layer was evaporated on the metal layers. The samples were annealed from 250 °C to 450 °C in a N₂ ambient for 1 h. Before performing XRD and 4-point probe measurements to investigate the properties of the α - or poly-Ge films, the upper SiO₂ capping layers were etched in 2% diluted hydrofluoric (HF) acid and the unreacted metal layers were removed by 36% hydrochloric

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(HCl) acid. The 4-point probe measurement was performed 3-times in each sample. In addition, TEM and SIMS analyses were performed on the following selected samples; (1) non-annealed and solid phase crystallized (SPC) sample annealed at 450 °C, (2) In-MIC sample annealed at 250 °C and 400 °C, and (3) Sn-MIC sample annealed at 400 °C. In the preparation of TLM samples, SPC and MIC samples were respectively annealed for an hour (at 450 °C) and 3 h (at 400 °C) to fully crystallize the 90 nm thick α -Ge films, followed by 50 nm thick In deposition process for contact electrodes. The distances between metal pads ($200\ \mu\text{m} \times 100\ \mu\text{m}$) in the TLM pattern were 100 μm , 120 μm , 160 μm , 240 μm , 320 μm , and 420 μm . The one-set of TLM measurement was performed 8-times in each sample at room temperature.

3. Results and discussion

In order to estimate the crystal quality of In- and Sn-MIC samples and compare it with SPC samples, XRD analysis was performed on the samples and full width half maximum (FWHM) values were plotted as a function of annealing temperature in Fig. 1, along with the XRD patterns (between 25° and 30°). Here, (111) peaks were used to analyze the degree of Ge crystal quality, because Ge (111) peaks show the strongest intensity than the others due to the lower energy state on the (111) surface [4]. Compared to the (111) peak of SPC sample which began to show up from 450 °C, the (111) peaks of In-MIC and Sn-MIC appeared at lower temperatures of 250 °C and 400 °C. Lower MIC point in In-Ge system originates from slightly lower eutectic temperature (156.29 °C) to that of Sn-Ge (231.1 °C) [10]. MIC samples have higher FWHM values above 0.68° than that (about 0.47°) of SPC sample, indicating a smaller Ge crystal grain. It is thought that sufficient supply of In and Sn into the Ge films causes a large number of heterogeneous nuclei which consequently results in smaller grain size after the re-crystallization process. Especially for In-MIC process, broader (111) peaks were observed below 400 °C,

meaning smaller crystal grains. As we have previously reported for different FWHM values on poly-Ge films obtained by metal-based MIC processes (Pd, Cu, Ni, Au, Co, Al, Pt, and Ti) [5], the difference in grain size after In- and Sn-MIC processes seems to be attributed from the different level of metal diffusion and subsequently different number of heterogeneous nuclei. In addition, FWHM values of In-MIC samples were reduced as a function of annealing temperature by continuous Ge crystal growth. In the case of Sn-MIC, although the similar reduction behavior in FWHM was observed when the annealing temperature increased from 400 °C to 450 °C, it is hard to mention that Ge crystals are continuously grown above 400 °C by Sn-MIC process because there is additional effect by self-nucleation process at 450 °C. We also note that the FWHM values extracted in the In-MIC samples annealed below 350 °C are not reliable because the Ge (111) peaks have too weak intensity. Based on another XRD measurements (not shown in this paper) performed by tilting the SPC, In-MIC, and Sn-MIC samples from 0° to 60°, we also note that FWHM values were slightly reduced in all samples after tilting (FWHM: 0.5° $\rightarrow$ 0.46° in SPC, 0.93° $\rightarrow$ 0.85° in In-MIC, and 0.72° $\rightarrow$ 0.65° in Sn-MIC). It consequently indicates that all of the samples are polycrystalline with small grains.

4-point probe measurements were performed 3-times in each sample to compare the sheet resistances of the SPC and MIC samples. Fig. 2 shows the sheet resistance values of SPC, In-MIC, and Sn-MIC samples as a function of annealing temperatures between 250 °C and 450 °C. Here, the deviations of each sheet resistance value are about 8–15%. Before the self-nucleation process which is known to occur between 400 °C and 450 °C [11], α -Ge film have $\sim 100\ \Omega/\text{sq}$ of sheet resistance values because of vacancy defects at the grain boundaries working as *p*-type dopants in Ge [12]. After 450 °C anneal, the sheet resistance increased up to 23 k Ω/sq because the number of vacancy defects was reduced as a result of grain size increased. The sheet resistance values comparable to those of 450 °C SPC sample were observed in the In-MIC samples and those showed a decreasing trend with higher annealing temperature (from 19 k Ω/sq at 250 °C to 1.8 k Ω/sq at 450 °C). It is thought that the increased thickness of crystallized region by the continuous In-MIC growth decreases the sheet resistance value as the annealing temperature increases. Interestingly, compared to the 450 °C SPC sample, comparable resistance values were observed after the Sn-MIC process although smaller crystal grains indicating higher number of vacancy defects and consequently lower sheet resistance (assuming the similar thickness of crystallized region) were achieved through the Sn-MIC process. This relatively high sheet resistance in the Sn-MIC samples seems to be originated from the fact that (1) Sn

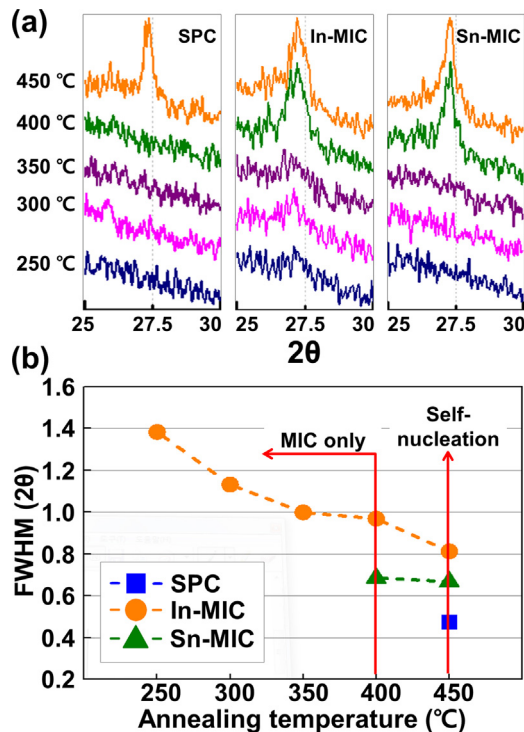


Fig. 1. (a) XRD patterns of SPC, In-MIC, and Sn-MIC samples annealed between 250 °C and 450 °C. (b) FWHM values extracted from (a) as a function of annealing temperature.

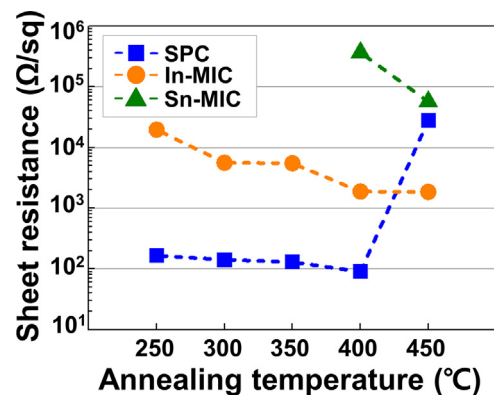


Fig. 2. Sheet resistances as a function of annealing temperature, which were measured on SPC, In-MIC, and Sn-MIC samples annealed between 250 °C and 450 °C.

works as a neutral atom in Ge and (2) bigger Ge crystal grains (less amount of grain boundary) are obtained by Sn-MIC, which lowers the resistivity reduction effect by Sn in Ge [13]. According to our previous report, it was confirmed that metals (Pd, Cu, Ni, Au, and Co) working as acceptors/donors and also causing the conduction through grain boundary in Ge helps to reduce the resistivity of poly-Ge after the MIC process [5].

Among the SPC and MIC samples, we then selected (1) non-annealed and solid phase crystallized (SPC) sample annealed at 450 °C, (2) In-MIC sample annealed at 250 °C and 400 °C, and (3) Sn-MIC annealed at 400 °C for TEM and SIMS analyses. Fig. 3 shows the cross-sectional TEM images including diffraction patterns taken on the selected samples and cross-sectional schematic diagram for the MIC samples. As expected, poly-Ge was confirmed after 450 °C SPC process by the TEM image and diffraction pattern as shown in Fig. 3(b). The size of crystal grains observed in the 450 °C SPC sample was 60–70 nm or bigger, which is similar to the

previously reported values [11]. Even though crystallization phenomenon occurred at much lower temperature (250 °C) for the In-MIC process, it was only limited to the surface region because the processing temperature was too low. Larger crystallized region of α -Ge was obtained after the annealing temperature was increased to 400 °C. However, the size of crystal grains was smaller than 50 nm due to large number of heterogeneous nuclei that were induced by In atoms. In our previous report, we also confirmed small crystal grains below 50 nm in the Ge films crystallized by Pd, Cu, and Au reacting with Ge at below 350 °C [5]. In addition, we observed an increase in interphase roughness (between α -Ge and poly-Ge) as the annealing temperature rose from 350 °C up to 400 °C. It is thought that the interphase roughness increase is attributed to the difference in crystal growth rates which are dependent on growth directions [7]. For Sn-MIC, crystallization phenomenon started at a relatively higher temperature (400 °C) and the rate of MIC (60–70 nm/h at 400 °C) was

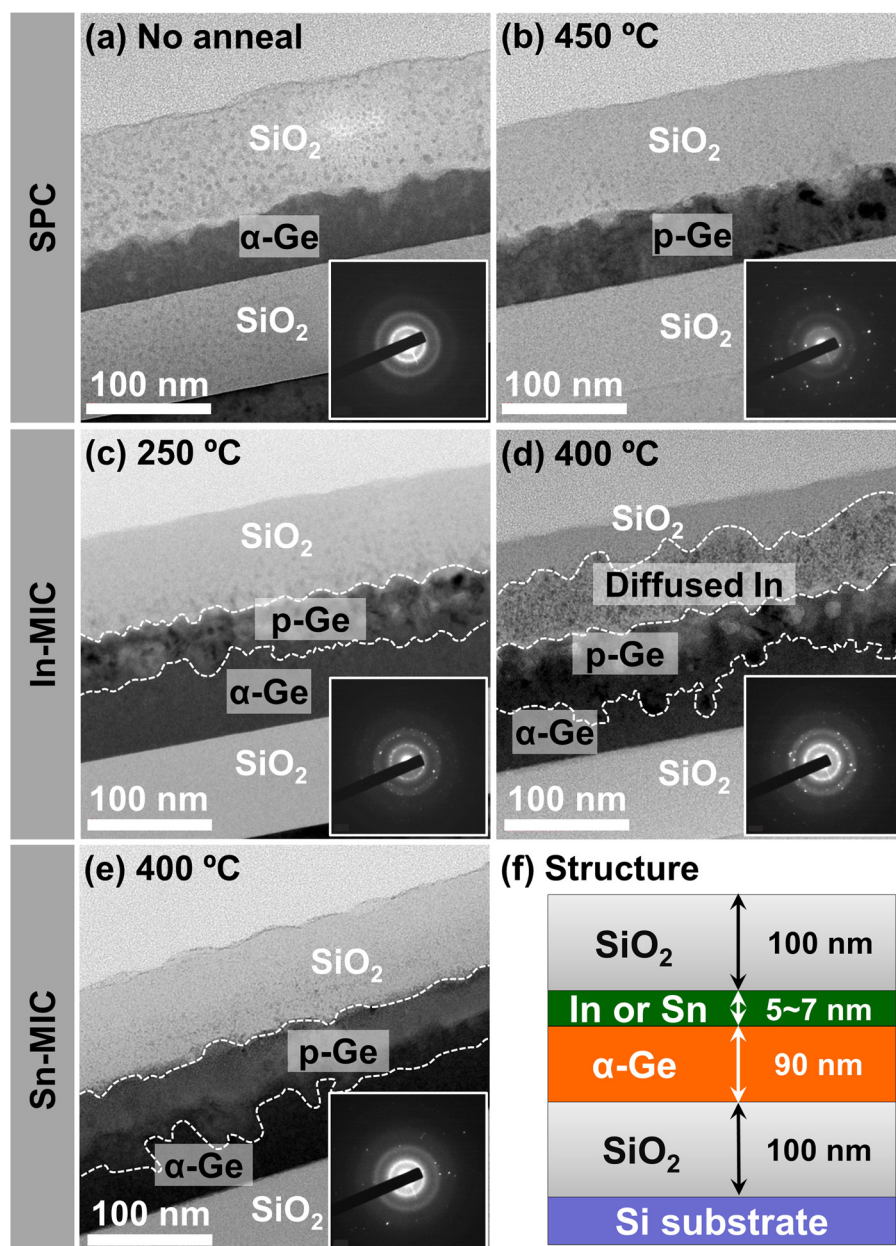


Fig. 3. Cross-sectional TEM Images and diffraction patterns of (a) SPC no annealed, (b) SPC 450 °C annealed, (c) In-MIC 250 °C annealed, (d) In-MIC 400 °C annealed, and (e) Sn-MIC 400 °C annealed samples. (f) Schematic diagram showing the MIC sample structure.

similar to the In-MIC process. In addition, compared to the In-MIC samples, bigger crystal grains of above 50 nm (correlated with the previous FWHM data) were obtained after the Sn-MIC process. Fig. 4 shows the SIMS profiles of Ge, In (or Sn), and SiO₂ elements as a function of sputtering time (zero sputtering time indicates a surface region) in the MIC structure. Here, peaks of metal (In or Sn) distributions exists between the upper SiO₂ and Ge even after 250 °C and 400 °C MIC process, because the metals that were initially supplied for the MIC process were too large in quantity to be consumed fully by the MIC process. As shown in the Fig. 4(a), In atoms diffused to the upper SiO₂ and Ge regions after the 250 °C anneal and about 46% of total In atoms are expected to move into the Ge region. At a higher annealing process (400 °C), this trend becomes higher (51%). Here, we calculated the percentage values by dividing the integrated metal signal (area) in the Ge region by the area in all regions. Sn atoms on the other hand, as seen in the

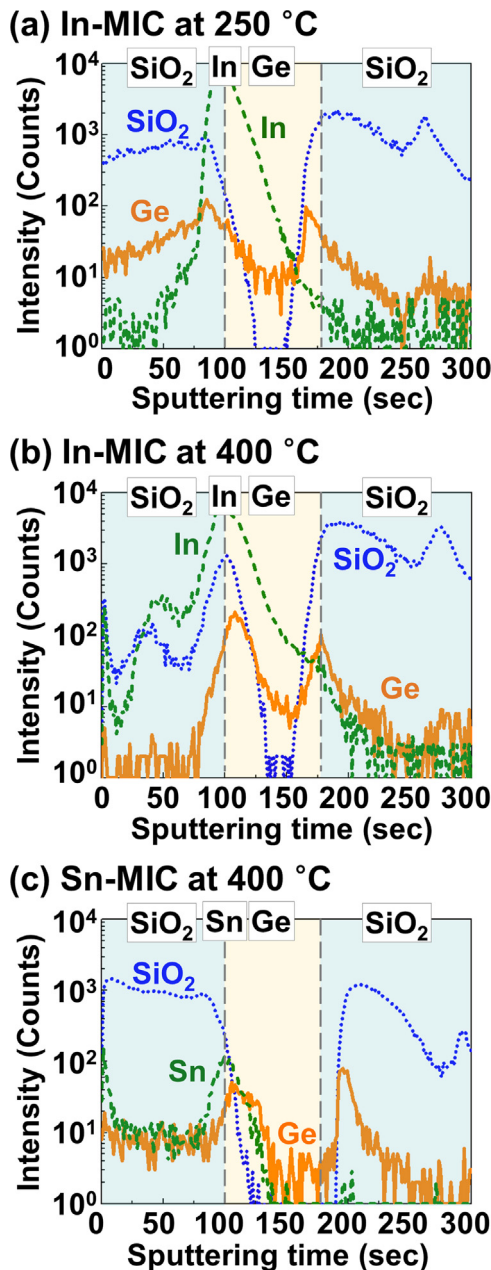


Fig. 4. SIMS profiles as a function of sputtering time, which were measured on (a) In-MIC 250 °C annealed, (b) In-MIC 400 °C annealed, and (c) Sn-MIC 400 °C annealed samples.

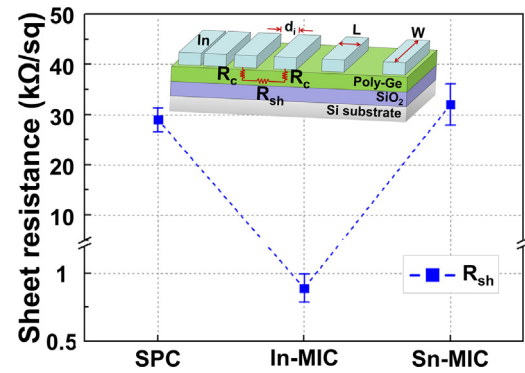


Fig. 5. Sheet resistances of SPC, In-MIC, Sn-MIC samples, which were fully crystallized. Inset: schematic diagram for TLM structure.

Fig. 4(c), diffuse much more slowly into the Ge region than In at 400 °C so that relatively lower Sn concentration is predicted in the Ge region during the anneal (only ~36% of total Sn atoms is expected to be located in the Ge). This consequently indicates that less number of heterogeneous nuclei was induced by Sn and it provided larger Ge crystal grains. The diffusion rate of Sn atoms into the Ge seems to be even slower than Sn diffusion in SiO₂.

TLM analysis was performed to compare the sheet resistances of the fully crystallized samples obtained by SPC and MIC techniques (Fig. 5). Based on the previous TEM images and the confirmation that the MIC rates in In- and Sn-MIC samples were about 60 nm/h at 400 °C, the samples were annealed at 400 °C for 3 h to fully crystallize. The sheet resistance values of fully crystallized MIC samples were slightly reduced from those of partially crystallized MIC samples (In-MIC: from 1.8 kΩ to 0.9 kΩ and Sn-MIC: from 57 kΩ to 31 kΩ) because of the increased thickness of the crystallized Ge regions. As previously confirmed, the Sn-MIC sample presented comparable resistance value to that of 450 °C SPC sample (due to neutral property Sn in Ge) and much higher than that of In-MIC sample. We also note that the measured contact resistance values are negligible because of (1) strong Fermi level pinning phenomenon of metals to about 0.1 eV above region from the valance band of Ge and (2) large number of defects in poly-Ge causing trap-assisted tunneling phenomenon [14].

4. Conclusion

In conclusion, In- and Sn-based MIC phenomenon on α-Ge was newly presented. The In- and Sn-MIC processes started at 250 °C and 400 °C, and larger crystallized regions were obtained after the In-MIC process as the temperature increased up to 400 °C. Compared to the In-MIC samples, higher sheet resistance (similar to that of SPC) and bigger crystal grains of above 50 nm (smaller than that of SPC) were obtained in the Sn-MIC samples. According to SIMS analysis, Sn atoms diffused much more slowly into the Ge region than In atoms at 400 °C, providing lower density of heterogeneous nuclei induced by metals and consequently larger crystal grains.

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